College of Engineering – Department of Electrical Engineering

CURRICULUM

Bachelor of Science in Electrical Engineering
Academic Year 2025-2026
Reference CMOs: CMO No. 88 s.2017, CMO No. 39, s. 2021, CMO No. 40, s. 2021 and CMO No. 20, s. 2013

Curriculum Description

The Bachelor of Science in Electrical Engineering program is designed to prepare students for careers as professional electrical engineers, researchers and/or for admission to graduate school who are expected to be engaged in relevant issues in society and industry. The program is structured to provide an optimal curriculum that affords graduates with firm foundations on the concepts and theories in electrical engineering through transformative learning strategies.

Graduates of this program are in academic, industrial, and government sectors. Career opportunities include power generation, power transmission, power distribution, instrumentation/automation, design and supervision, construction and management, corporate management, sales and marketing, information and computing technology and academe.

Program Education Objectives

The electrical engineering alumni three to five years after graduation shall:

PEO 1: Demonstrate professional expertise through sustainable, analytical and innovative thinking for the purpose of solving industry-based engineering problems, involvement in research-oriented projects and adapting to the emerging trends in the advancements in electrical engineering field.

PEO 2: Exhibit holistic leadership and professionalism, through peer-recognized expertise in a team-based environment as agents of sustainable economic development.

PEO 3: Engage the Electrical Engineering profession in the globally competitive environment through continuous quality education, research and development, and other creative and innovative efforts in science, engineering and technology, as well as other professional careers.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex electrical engineering problems;

SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability

issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex electrical engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex electrical engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex electrical engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

College of Engineering – Department of Electrical Engineering
CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Probability and Statistics for AI	3	0	3
Discrete Mathematics	3	0	3
Subtotal	**17**	**0**	**17**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers 1	3	3	4
Physics for Engineers 2	2	3	3
Subtotal	**8**	**9**	**11**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Engineering Economics	3	0	3
Technopreneurship	3	0	3
Engineering Mechanics	3	0	3
Fundamentals of Deformable Bodies	3	0	3
Fluid Mechanics	3	0	3
Environmental Science and Engineering	3	0	3
Basic Occupational Safety and Health	3	0	3
Materials Science and Engineering	3	0	3
Subtotal	**24**	**9**	**27**
D. Allied Courses			
Electronic Circuits: Devices and Analysis	3	3	4
Basic Thermodynamics	2	0	2
Industrial Electronics	3	3	4
Fundamentals of Electromagnetics	2	0	2
Fundamentals of Electronic Communications	3	0	3
Logic Circuits and Switching Theory	2	3	3
Microprocessor Systems	3	0	3
Computer Programming	0	3	1
Subtotal	**18**	**12**	**22**
E. Professional Courses			
Methods of Research	3	0	3
On-the-Job Training	320 hours		4
Instrumentation and Control	2	3	3
Electrical Circuits 1	3	3	4
Advanced Mathematics for EE	3	0	3
Electrical Circuits 2	4	3	5
Electrical Machines 1	2	3	3
Electrical Apparatus and Devices	2	3	3
Electrical Machines 2	3	3	4
EE Laws, Contacts and Professional Ethics	2	0	2

College of Engineering – Department of Electrical Engineering

Feedback Control System	3	0	3
Electrical Standards and Practices	0	3	1
Power System Analysis	4	3	5
Management of Engineering Projects	2	0	2
Numerical Methods and Analysis for EE	2	3	3
EE Design Project 1	0	3	1
Electrical Systems and Illumination Design	3	6	5
Fundamentals of Power Plant Engineering Design	2	3	3
Transmission, Distribution Systems and Substation Design	2	3	3
Electrical Engineering Practice with Comprehensive Examination	0	6	2
Electrical Maintenance and Operations	3	0	3
Seminars/Colloquia	0	3	1
EE Design Project 2	0	3	1
Subtotal	**45**	**54**	**67**
F. Elective Courses			
EE Elective 1	3	0	3
EE Elective 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL TECHNICAL COURSES	**118**	**84**	**150**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives and Mandated Courses			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	2	3	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3

College of Engineering – Department of Electrical Engineering

Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**49**	**3**	**50**
GRAND TOTAL	**167**	**87**	**200**

SUMMARY

Courses	**Number of Units**
I. TECHNICAL COURSES	
A. Mathematics	**17**
B. Natural/Physical Sciences	**11**
C. Basic Engineering Sciences	**27**
D. Allied Courses	**22**
E. Professional Courses	**67**
F. Elective Courses	**6**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives and Mandated Courses	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)	**8**
D. National Service Training Program	**6**
GRAND TOTAL	**200**

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College of Engineering – Department of Electrical Engineering
CURRICULUM STRUCTURE

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CHEM 101	Chemistry for Engineers	4	3	3			TC-B
CpE 101	Computer Programming	1	0	3			TC-D
ENGG 101	Introduction to Engineering	1	0	3			TC-C
GEd 104	The Contemporary World	3	3	0			NTC-A
GEd 105	Readings in Philippine History	3	3	0			NTC-A
GEd 108	Art Appreciation	3	3	0			
MATH 101	Differential Calculus	4	4	0			TC-A
PATHFIT 1	Movement Competency Training	2	2	0			NTC-C
NSTP 111	National Service Training Program 1	3	3	0			NTC-D
	Total	**24**	**21**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 102	Engineering Drawing	1	0	3			TC-C
GEd 101	Understanding the Self	3	3	0			NTC-A
GEd 102	Mathematics in the Modern World	3	3	0			NTC-A
GEd 106	Purposive Communication	3	3	0			NTC-A
GEd 109	Science, Technology and Society	3	3	0			NTC-A
MATH 102	Integral Calculus	4	4	0	MATH 101		TC-A
PHYS 102	Physics for Engineers 1	4	3	3	MATH 101	MATH 102	TC-B
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC-C
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC-D
	Total	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
PHYS 103	Physics for Engineers 2	3	2	3	PHYS 102,MATH 102		TC-B
GEd 103	Life and Works of Rizal	3	3	0			NTC-B
GEd 107	Ethics	3	3	0			NTC-A
	TOTAL	**9**	**8**	**3**			

SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 101	Linear Algebra for AI	3	3	0			NTC-B
CpE 202	Discrete Mathematics	3	3	0	MATH 102		TC-A
ECE 413	Fundamentals of	2	2	0	PHYS 103	MATH 103	TC-D

College of Engineering – Department of Electrical Engineering

	Electromagnetics						
EE 201	Electrical Circuits 1	4	3	3	MATH 102, PHYS 103		TC-E
ENGG 109	Engineering Mechanics	3	3	0	PHYS 102		TC-C
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102		TC-C
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
ME 210	Basic Thermodynamics	2	2	0	PHYS 103, MATH 102		TC-D
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1, PATHFit 2		NTC-C
	Total	**23**	**21**	**6**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC-A
ECE 405	Electronic Circuits: Devices and Analysis	4	3	3	EE 201		TC-D
EE 202	Advanced Mathematics for EE	3	3	0	MATH 103		TC-E
EE 203	Electrical Circuits 2	5	4	3	EE 201	EE 202	TC-E
EE 204	Electrical Machines 1	3	2	3	EE 201, ECE 413		TC-E
ENGG 110	Fundamentals of Deformable Bodies	3	3	0	ENGG 109		TC-C
ENGG 114	Basic Occupational Safety and Health	3	3	0			TC-C
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	2	0	PATHFit 1, PATHFit 2		NTC-C
	Total	**26**	**23**	**9**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 301	Logic Circuits and Switching Theory	3	2	3	ECE 405		TC-D
ECE 414	Fundamentals of Electronic Communications	3	3	0	ECE 405		TC-D
EE 301	Electrical Apparatus and Devices	3	2	3	EE 203, EE 204		TC-E
EE 302	Electrical Machines 2	4	3	3	EE 203, EE 204	EE 301	TC-E
EE 303	EE Laws, Contacts and Professional Ethics	2	2	0	3rd year standing		TC-E
EE 304	Feedback Control System	3	3	0	EE 202, ECE 405		TC-E
ENGG 104	Engineering Economics	3	3	0	AI 102		TC-C
ENGG 116	Methods of Research	3	3	0	AI 102, 3rd year standing		TC-E
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101		TC-C
	Total	**27**	**24**	**9**			

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THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 103	Machine Learning and Neural Networks	3	2	3	AI 101, AI 102, CpE 101		NTC-B
CpE 307	Microprocessor Systems	3	3	0	CpE 301		TC-D
ECE 407	Industrial Electronics	4	3	3	ECE 405		TC-D
EE 305	Electrical Standards and Practices	1	0	3	EE 303		TC-E
EE 306	Power System Analysis	5	4	3	EE 301, EE 302	EE 305	TC-E
ENGG 112	Fluid Mechanics	3	3	0	ENGG 110, PHYS 103		TC-C
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		TC-C
GEdE 101	The Entrepreneurial Mind	3	3	0			NTC-B
	Total	**25**	**21**	**12**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
EE 307	Management of Engineering Projects	2	2	0	ENGG 104		TC-E
EE 308	Numerical Methods and Analysis for EE	3	2	3	MATH 103, CpE 101		TC-E
EEE 401	EE Elective 1	3	3	0	3rd year standing		TC-F
	Total	**8**	**7**	**3**			

FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
EE 401	EE Design Project 1	1	0	3	ENGG 116, 4th Year Standing		TC-E
EE 402	Electrical Systems and Illumination Design	5	3	6	EE 306		TC-E
EEE 402	EE Elective 2	3	3	0	EEE 401		TC-F
ENGG 117	On-the-Job Training	4	320		4th year standing		TC-E
	Total	**13**	**6**	**9**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
EE 403	Fundamentals of Power Plant Engineering Design	3	2	3	EE 306		TC-E
EE 404	Transmission, Distribution Systems and Substation Design	3	2	3	EE 306		TC-E
EE 405	Electrical Engineering Practice with Comprehensive Examination	2	0	6	Graduating		TC-E
EE 406	Electrical Maintenance and Operations	3	3	0	4th Year Standing		TC-E

College of Engineering – Department of Electrical Engineering

EE 407	Seminars/Colloquia	1	0	3	Regular Standing		TC-E
EE 408	EE Design Project 2	1	0	3	Graduating		TC-E
ENGG 105	Technopreneurship	3	3	0	4th year standing		TC-C
ICE 420	Instrumentation and Control	3	2	3	EE 304		TC-E
	Total	**19**	**12**	**21**			

TOTAL UNITS: 200

EE Elective Tracks

Renewable Energy for Sustainable Development

Solar Energy
Biogas/Biomass Energy

Machine Automation and Process Control

Pneumatic and Electropneumatics
Programmable Logic Controlled in Manufacturing & Power Systems